

Properties

Real Number Properties

Let a , b , and c be any real numbers.

Addition Property of Equality	If $a = b$, then $a + c = b + c$ and $c + a = c + b$.	
Addition Property: Addends with Like Signs	For all $a > 0$ and $b > 0$, $a + b = a + b $. For all $a < 0$ and $b < 0$, $a + b = - a + b $.	
Addition Property: Addends with Unlike Signs	For all $a > 0$ and $b < 0$, If $ a > b $, then $a + b = a - b $. If $ a < b $, then $a + b = - b + a $.	
Subtraction Property of Equality	If $a = b$, then $a - c = b - c$.	
Substitution Property of Equality	If $a = b$, then a may be replaced with b in any expression or equation.	
Multiplication Property of Equality	If $a = b$, then $c \cdot a = c \cdot b$ and $a \cdot c = b \cdot c$.	
Division Property of Equality	If $a = b$ and $c \neq 0$, then $\frac{a}{c} = \frac{b}{c}$.	
Distributive Property	$a(b + c) = ab + ac$	
	Addition	Multiplication
Commutative Properties	$a + b = b + a$	$a \cdot b = b \cdot a$
Associative Properties	$(a + b) + c = a + (b + c)$	$(a \cdot b) \cdot c = a \cdot (b \cdot c)$
Inverse Properties	$a + (-a) = 0$ and $(-a) + a = 0$	$a \cdot \frac{1}{a} = 1$ and $\frac{1}{a} \cdot a = 1, a \neq 0$
Identity Properties	$a + 0 = a$ and $0 + a = a$	$a \cdot 1 = a$ and $1 \cdot a = a$

Absolute Value Equations

If $|x| = a$ for some positive number a , then $x = a$ or $x = -a$.

Properties of Exponents

Let a and b be nonzero real numbers. Let m and n be integers.

If n is a positive integer, then $a^n = a \cdot a \cdot a \cdot \dots \cdot a$ (n factors).

Zero Exponent Property	$a^0 = 1, a \neq 0$
Negative Exponent Property	$a^{-m} = \frac{1}{a^m}, a \neq 0$
Product of Powers Property	$a^m \cdot a^n = a^{m+n}$

Square Root Properties

For nonnegative values of m , n , and p , if $m < n < p$, then $\sqrt{m} < \sqrt{n} < \sqrt{p}$.

Product Property	For real numbers a and b , $\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$ and $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$.
Quotient Property	For real numbers a and b with $b \neq 0$, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$.

Reciprocal Properties

Reciprocal Property of Multiplication	For any nonzero real number a , $a \cdot \frac{1}{a} = 1$.
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For all nonzero real numbers a and b , the reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$.

For any nonzero real number a , $\frac{1}{-a} = \frac{-1}{a} = -\frac{1}{a}$.

For all nonzero real numbers a and b , $\frac{1}{ab} = \frac{1}{a} \cdot \frac{1}{b}$.

Division Properties

For any real number a and nonzero real number b , $a \div b = a \cdot \frac{1}{b}$.

For all real numbers a and b and nonzero real number c , $a + \frac{b}{c} = \frac{a}{c} + \frac{b}{c}$.

For all $a > 0$ and $b > 0$, $a \div b > 0$.

For all $a < 0$ and $b < 0$, $a \div b > 0$.

For all $a < 0$ and $b > 0$, $a \div b < 0$.

Properties of Order

Comparison Property of Order	If $a > b$, then $b < a$. If $a < b$, then $b > a$.
Transitive Property of Order	If $a > b$ and $b > c$, then $a > c$. If $a < b$ and $b < c$, then $a < c$.
Addition Property of Order	If $a > b$, then $a + c > b + c$. If $a < b$, then $a + c < b + c$.
Subtraction Property of Order	If $a > b$, then $a - c > b - c$. If $a < b$, then $a - c < b - c$.
Multiplication Property of Order, Positive Multiplier	If $a > b$ and $c > 0$, then $ca > cb$ and $ac > bc$. If $a < b$ and $c > 0$, then $ca < cb$ and $ac < bc$.
Multiplication Property of Order, Negative Multiplier	If $a > b$ and $c < 0$, then $ca < cb$ and $ac < bc$. If $a < b$ and $c < 0$, then $ca > cb$ and $ac > bc$.
Division Property of Order, Positive Multiplier	If $a > b$ and $c > 0$, then $\frac{a}{c} > \frac{b}{c}$. If $a < b$ and $c > 0$, then $\frac{a}{c} < \frac{b}{c}$.
Division Property of Order, Negative Multiplier	If $a > b$ and $c < 0$, then $\frac{a}{c} < \frac{b}{c}$. If $a < b$ and $c < 0$, then $\frac{a}{c} > \frac{b}{c}$.

Comparison Property of Rational Numbers

For nonzero integers a and c and positive integers b and d ,

$\frac{a}{b} > \frac{c}{d}$ if, and only if, $ad > bc$.

$\frac{a}{b} < \frac{c}{d}$ if, and only if, $ad < bc$.

Properties of Proportions

Let a , b , c , and d be real numbers.

Means-Extremes Product Property	$\frac{a}{b} = \frac{c}{d}$ if, and only if, $ad = bc$, given that b and d are not 0.
Reciprocal Property	If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$, given that a , b , c , and d are all nonzero.